BACHELOR OF COMPUTER APPLICATIONS

# KLE TECHNOLOGICAL UNIVERSITY

B. V. Bhoomaraddi Campus

# Tech Cart

**By**

**Team: Code Companions**

| Name | USN |
|---|---|
| Vaishnavi L Kulkarni | 01FE23BCA237 |
| Arundati K Huragi | 01FE23BCA236 |
| Sneha S Kittur | 01FE23BCA193 |

**Under the guidance of**

**Prof. Hemalatha J P**

**KLE TECHNOLOGICAL UNIVERSITY**

**B. V. Bhoomaraddi College**

**Vidyanagar Hubballi-580031**

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# SOFTWARE REQUIREMENT SPECIFICATION

## 1.0 INTRODUCTION

The purpose of this document is to define the software requirements for the Online Electronics Shopping Cart, an E-Commerce web application designed to sell computer science-related electronics products online and to allow store managers or administrators to manage inventory. Users can browse, select, and add to the cart with this online shopping cart and system configurations.

### 1.1 Scope

- Enable browsing and purchasing of electronic products online.
- User registration, login, profile management, and order tracking.
- Real-time cart updates and inventory management.
- Secure payment gateway integration and order confirmation via email.
- Admin functionalities like product management, order handling, and user feedback management.

## 2.0 OVERALL DESCRIPTION

The Online Electronics Shopping Cart system is an e-commerce web application developed to simplify the online purchase of electronic items. It provides a seamless shopping experience for customers and a robust management platform for administrators.

### Key Features & Functions

**Admin Side:**

- Admins can add, update, or delete products and maintain the catalog.
- Inventory management with real-time stock updates and low-stock alerts.
- Order management for monitoring, updating, and processing customer orders
- Analytics dashboard to view sales reports, order statistics, and stock levels.
- Manage registered users and handle customer queries/feedback.

**Customer Side:**

- Users can register, log in, and manage profiles.
- Products are displayed with categories, descriptions, prices, and images.

- Advanced search and filter options (by price, brand, rating).
- A shopping cart allows adding, updating, or removing products in real-time.
- Checkout is supported with secure payment options such as UPI, net banking, and cards
- Customers can view their order history and receive automated email/SMS notifications about order status (placed, shipped, delivered).

**System Characteristics**

- **Architecture:** Multi-tier (Presentation, Business Logic, Database).
- **Technology Stack:** Java, JSP, JDBC, MySQL (backend); HTML, CSS, JavaScript (frontend); deployed on Tomcat server.
- **Responsiveness:** Optimized for desktops, tablets, and smartphones.
- **Security:** Encrypted login, secure payment gateway integration, and protection against cyberattacks.
- **Scalability:** Designed to handle a growing number of users, products, and transactions.
- **Accessibility:** Inclusive design for users with disabilities (screen readers, high-contrast support).

# 3.0 SPECIFIC REQUIREMENTS

## 3.1 Functional Requirements

### 3.1.1 User Module

- User shall register with email, username, and password.
- User shall login and logout securely.
- User can browse, search, and filter electronics products.
- User can add items to cart, update quantity, and remove items.
- User can checkout and process payments.
- User can view order history and track shipping status.
- User can provide feedback and update profile details.

### 3.1.2 Admin Module

- Admin can add, update, and delete product listings.
- Admin can manage stock and inventory.
- Admin can view and process orders (ship/unship).
- Admin can view user feedback and manage user accounts.

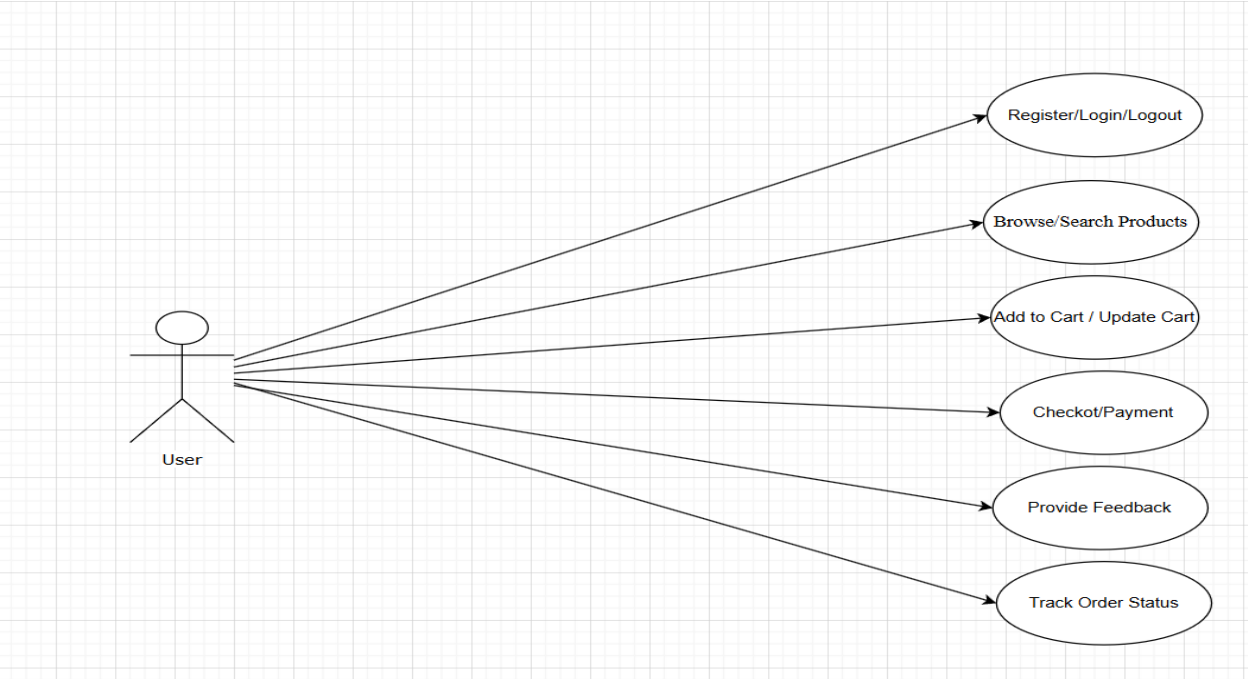
### 3.2 Non-Functional Requirements

- The system should handle at least 100 concurrent users.
- Cart updates should reflect in real-time.
- User passwords should be encrypted.
- Secure payment gateway integration.
- Access controls for admin and regular users.
- Responsive and user-friendly UI across devices.
- Minimal clicks to complete a purchase.
- Clear error messages and form validations.

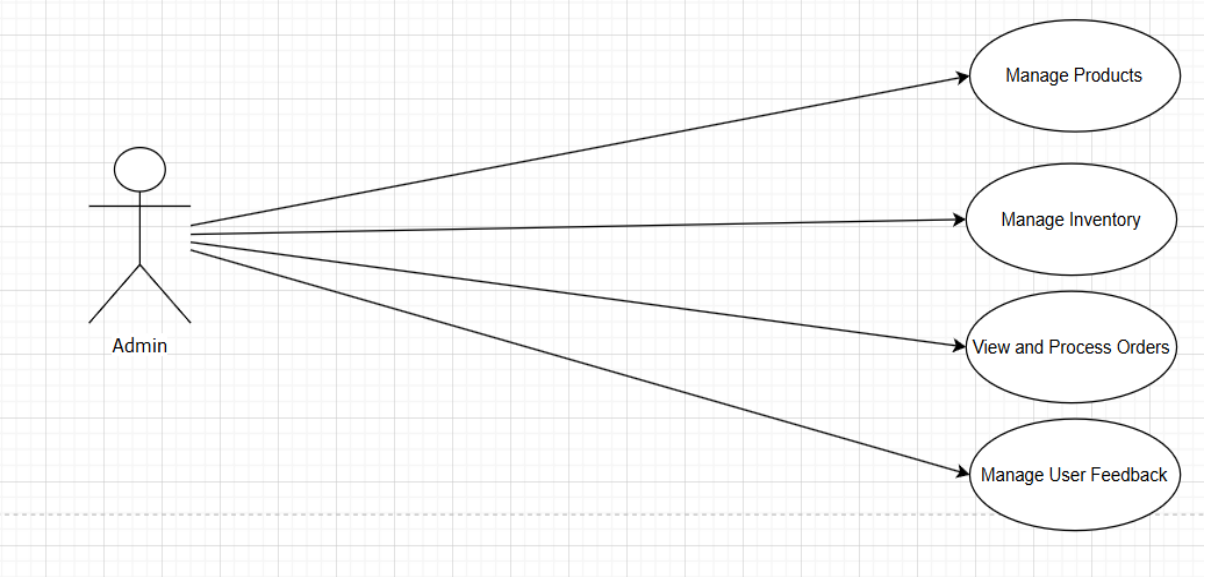
## 4.0 SYSTEM MODELS

### 4.1 Use Case Diagram

**User:**



## Admin:


Admin
Manage Products
Manage Inventory
View and Process Orders
Manage User Feedback